Exercise 3B

1) Force causes motion
2) No
3) As the ball ~~rolls~~ starts moving, frictional force starts to act on it. therefore stoping it.

7 a)

According to Newton's first law of motion, An object will continue in its state of motion until and unless acted upon by a non-zero net-force. Now what this statement states that is an object which can be anything will continue moving or resting until a force like a push is applied to change its state of motion

9) In the first Law of Motion, Newton says that an object cannot change its state of motion on its own. It needs a force for that. So if an object is in rest or motion it will stay like that only until some non-zero net force.

10) Inertia mainly depends on mass as mass ~ inertia which means that more the mass greater the inertia

12) Inertia is the resistance to have a change in motion / rest. And as we know mass ~ inertia which means that if we had more mass it would have more inertia which means it would be harder to move the body from rest

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(14)

(i) When a bike suddenly accelerates you tend to lean back and when it deaccelerate you tend to lean forward

(ii) A Bowler runs before bowling the cricket ball so that he the ball can continue with that force + the force of the throw which makes it go faster.

15) No, because the forces cancel out each other. And it's clearly stated that the force needed to move an object should be non-zero net force.

17 (a) Ans : Net force = Zero

19) Its because of the coins inertia. So when the card is quickly flicked off, the resistance to going from rest to motion of the coin is so much that it just stays in rest while the card is flicked of and falls down into the tumbler due to gravity.

20) Because the ball is moving relative to the train which mean because the the boy is moving with the ball too.

21)

(i) When the train suddenly accelerates, the passenger tends to fall back due to his inertia or his resistance to change in motion.

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1e) Just as the above the sliding doors have their own resistance to go from rest to motion and therefore tend to open sometimes

(r) When you shake the branch the fruits due to their own tend to resist motion which in turn causes their attachment to the branch to loser, therefore making itself falldown.

(f) (a) When the carpet is suddenly shaken due the inertia of dust particle, the particle tends to stay in rest and inturn falls down due to gravity therefore moving outof the carpet.

(d) When you get out of a moving bus you tend to continue in your state of motion but if you stop, the force of inertia will easily overpower you.

(f) It helps because when you are running your inertia is in motion and that will be advantageous for some extra force to jump further.

MCQs

1) (b) 3) (a)
2) (b)

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